Class11 Alphafold

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Background

In this hands-on session we will utilize AlphaFold to predict protein structure from sequence (Jumper et al. 2021).

Without the aid of such approaches, it can take years of expensive laboratory work to determine the structure of just one protein. With AlphaFold we can now accurately compute a typical protein structure in as little as ten minutes.

The PDB database (the main repository of experimental structures) only has ~**250 thousand structures** (we saw this in the last lab) . The main protein sequence database has over **200 million** sequences ! Only 0.125% of known sequences have a known structure - this is called the "structure knowledge gap".

Structures are much harder to determine than sequences. They are expensive (on average ~$1 million each) They take on average 3-5 years to solve!

EBI Alphafold Database

The EBI has a database of pre-computed AlphaFold (AF) modules called AFDB. This is growing all the time and can be useful to check before running AF ourselves. ## Running Alphafold

We can download and run locally (on our own computers) but we need a GPU. Or we can use "cloud" computing to run this on someone elses computers

We will use ColabFold < https://github.com/sokrypton/ColabFold >

We previously found that there was no AFDB entry for our HIV sequence:

```
>HIV-Pr-Dimer
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQ
```

We will use AlphaFold2_mmseqs2

8 Custom analysis of resulting models

```
results_dir <- "hivpr_23119"
```

```
pdb_files <- list.files(path=results_dir,
                        pattern="*.pdb",
                        full.names = TRUE)


basename(pdb_files)
```

```
[1] "hivpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb"
[2] "hivpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb"
[3] "hivpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb"
[4] "hivpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb"
[5] "hivpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb"
```

```
library(bio3d)

# Read all data from Models
#  and superpose/fit coords
pdbs <- pdbaln(pdb_files, fit=TRUE, exefile="msa")
```

```
Reading PDB files:
hivpr_23119/hivpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb
hivpr_23119/hivpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb
hivpr_23119/hivpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb
hivpr_23119/hivpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb
hivpr_23119/hivpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb
.....

Extracting sequences

pdb/seq: 1   name: hivpr_23119/hivpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4
pdb/seq: 2   name: hivpr_23119/hivpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1
pdb/seq: 3   name: hivpr_23119/hivpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5
pdb/seq: 4   name: hivpr_23119/hivpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2
pdb/seq: 5   name: hivpr_23119/hivpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3
```

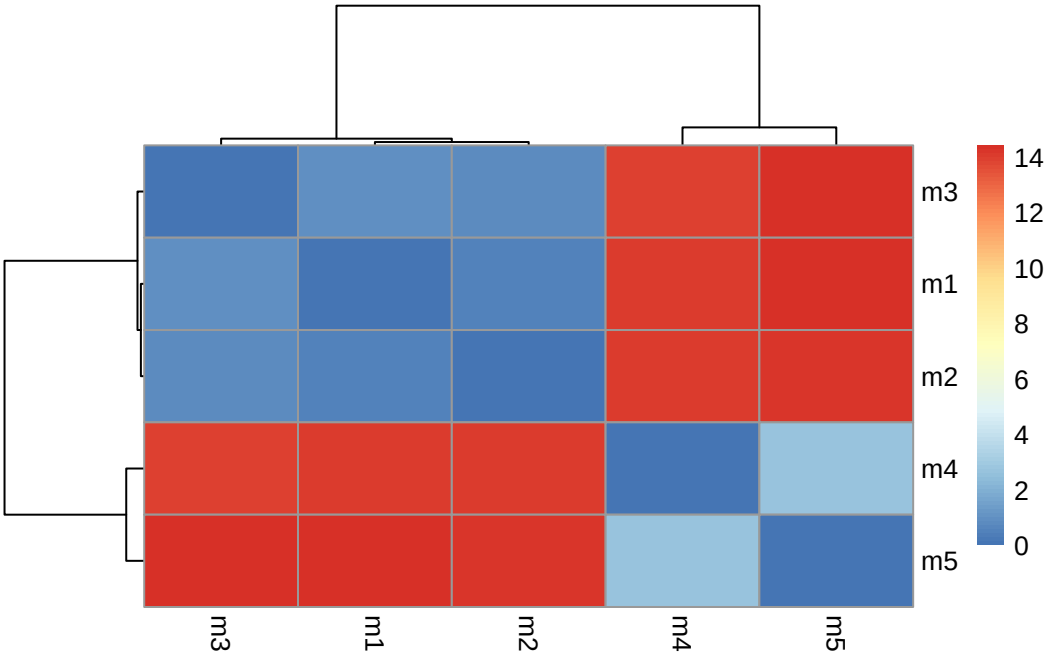
```r
rd <- rmsd(pdbs, fit=T)
```

```
Warning in rmsd(pdbs, fit = T): No indices provided, using the 198 non NA positions
```

```r
range(rd)
```

```
[1]  0.000 14.428
```

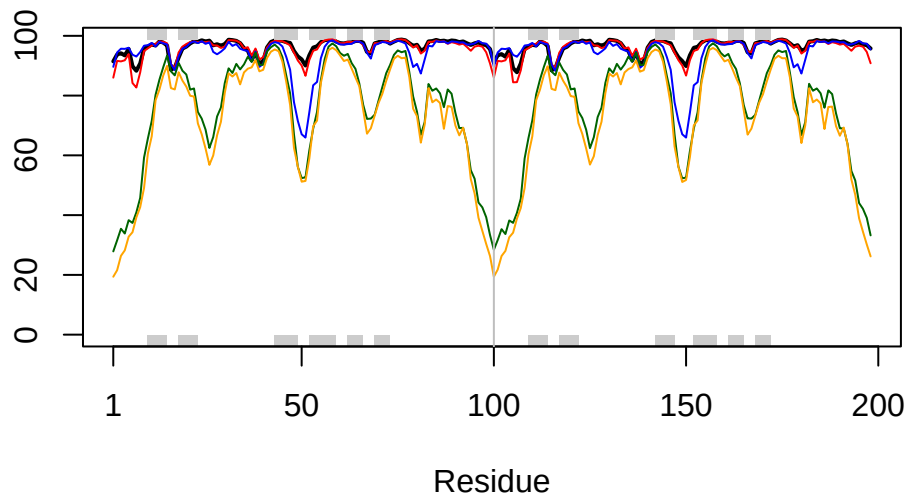
```r
library(pheatmap)

colnames(rd) <- paste0("m",1:5)
rownames(rd) <- paste0("m",1:5)
pheatmap(rd)
```




```r
pdb <- read.pdb("1hsg")
```

```
  Note:  Accessing on-line PDB file
```

```r
plotb3(pdbs$b[1,], typ="l", lwd=2, sse=pdb)
points(pdbs$b[2,], typ="l", col="red")
points(pdbs$b[3,], typ="l", col="blue")
points(pdbs$b[4,], typ="l", col="darkgreen")
points(pdbs$b[5,], typ="l", col="orange")
abline(v=100, col="gray")
```




```r
core <- core.find(pdbs)
```

```
core size 197 of 198  vol = 8545.079
core size 196 of 198  vol = 7894.973
core size 195 of 198  vol = 3576.564
core size 194 of 198  vol = 1851.077
core size 193 of 198  vol = 1697.205
core size 192 of 198  vol = 1612.775
core size 191 of 198  vol = 1530.208
core size 190 of 198  vol = 1447.408
core size 189 of 198  vol = 1377.117
core size 188 of 198  vol = 1303.826
core size 187 of 198  vol = 1239.04
core size 186 of 198  vol = 1188.139
core size 185 of 198  vol = 1118.473
```

```
core size 184 of 198  vol = 1071.674
core size 183 of 198  vol = 1034.016
core size 182 of 198  vol = 980.836
core size 181 of 198  vol = 942.246
core size 180 of 198  vol = 911.395
core size 179 of 198  vol = 879.756
core size 178 of 198  vol = 834.466
core size 177 of 198  vol = 785.268
core size 176 of 198  vol = 762.122
core size 175 of 198  vol = 722.029
core size 174 of 198  vol = 700.395
core size 173 of 198  vol = 677.257
core size 172 of 198  vol = 657.81
core size 171 of 198  vol = 632.907
core size 170 of 198  vol = 614.198
core size 169 of 198  vol = 591.516
core size 168 of 198  vol = 573.979
core size 167 of 198  vol = 552.403
core size 166 of 198  vol = 529.489
core size 165 of 198  vol = 500.545
core size 164 of 198  vol = 482.517
core size 163 of 198  vol = 458.426
core size 162 of 198  vol = 444.455
core size 161 of 198  vol = 433.581
core size 160 of 198  vol = 419.086
core size 159 of 198  vol = 404.934
core size 158 of 198  vol = 393.803
core size 157 of 198  vol = 383.003
core size 156 of 198  vol = 366.654
core size 155 of 198  vol = 352.026
core size 154 of 198  vol = 335.663
core size 153 of 198  vol = 319.398
core size 152 of 198  vol = 307.935
core size 151 of 198  vol = 296.818
core size 150 of 198  vol = 284.289
core size 149 of 198  vol = 273.459
core size 148 of 198  vol = 261.978
core size 147 of 198  vol = 249.6
core size 146 of 198  vol = 237.954
core size 145 of 198  vol = 226.1
core size 144 of 198  vol = 213.265
core size 143 of 198  vol = 200.214
core size 142 of 198  vol = 187.504
```

```
core size 141 of 198  vol = 177.525
core size 140 of 198  vol = 167.372
core size 139 of 198  vol = 160.875
core size 138 of 198  vol = 154.455
core size 137 of 198  vol = 148.439
core size 136 of 198  vol = 142.13
core size 135 of 198  vol = 136.529
core size 134 of 198  vol = 130.77
core size 133 of 198  vol = 123.868
core size 132 of 198  vol = 117.609
core size 131 of 198  vol = 112.71
core size 130 of 198  vol = 106.361
core size 129 of 198  vol = 100.591
core size 128 of 198  vol = 95.718
core size 127 of 198  vol = 91.068
core size 126 of 198  vol = 86.862
core size 125 of 198  vol = 82.309
core size 124 of 198  vol = 78.554
core size 123 of 198  vol = 74.632
core size 122 of 198  vol = 70.489
core size 121 of 198  vol = 66.802
core size 120 of 198  vol = 62.901
core size 119 of 198  vol = 59.152
core size 118 of 198  vol = 55.75
core size 117 of 198  vol = 51.832
core size 116 of 198  vol = 48.3
core size 115 of 198  vol = 44.927
core size 114 of 198  vol = 42.418
core size 113 of 198  vol = 39.425
core size 112 of 198  vol = 37.381
core size 111 of 198  vol = 33.06
core size 110 of 198  vol = 28.153
core size 109 of 198  vol = 25.33
core size 108 of 198  vol = 22.509
core size 107 of 198  vol = 20.695
core size 106 of 198  vol = 18.754
core size 105 of 198  vol = 17.757
core size 104 of 198  vol = 16.712
core size 103 of 198  vol = 15.44
core size 102 of 198  vol = 14.745
core size 101 of 198  vol = 14.758
core size 100 of 198  vol = 13.11
core size 99 of 198  vol = 11.018
```

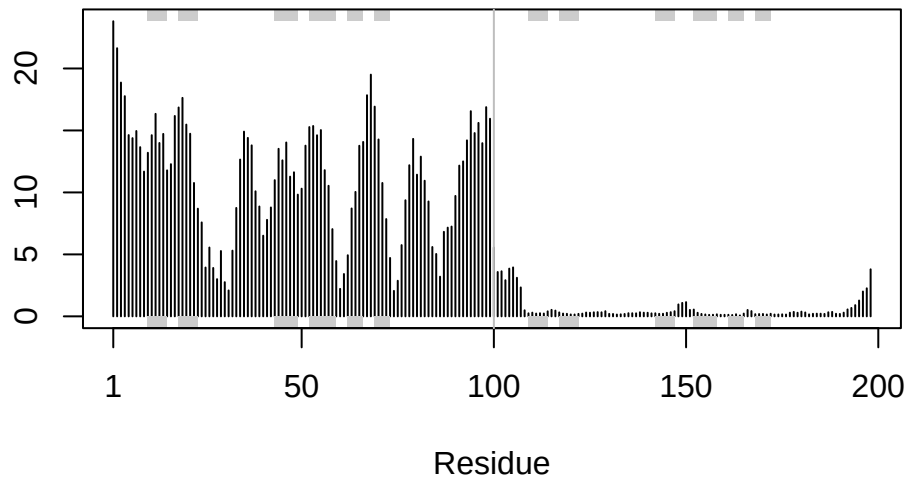
```
 core size 98 of 198  vol = 8.967
 core size 97 of 198  vol = 7.643
 core size 96 of 198  vol = 6.326
 core size 95 of 198  vol = 5.37
 core size 94 of 198  vol = 4.312
 core size 93 of 198  vol = 3.391
 core size 92 of 198  vol = 2.697
 core size 91 of 198  vol = 1.911
 core size 90 of 198  vol = 1.577
 core size 89 of 198  vol = 1.144
 core size 88 of 198  vol = 0.826
 core size 87 of 198  vol = 0.594
 core size 86 of 198  vol = 0.494
 FINISHED: Min vol ( 0.5 ) reached
```

```r
core.inds <- print(core, vol=0.5)
```

```
# 87 positions (cumulative volume <= 0.5 Angstrom^3)
  start end length
1     8  50     43
2    52  95     44
```

```r
xyz <- pdbfit(pdbs, core.inds, outpath="corefit_structures")
```

```r
rf <- rmsf(xyz)

plotb3(rf, sse=pdb)
abline(v=100, col="gray", ylab="RMSF")
```



9 Predicted Alignment Error for domains

```
library(jsonlite)

pae_files <- list.files(path=results_dir,
                        pattern=".*model.*\\.json",
                        full.names = TRUE)
```

```
pae1 <- read_json(pae_files[1],simplifyVector = TRUE)
pae5 <- read_json(pae_files[5],simplifyVector = TRUE)

attributes(pae1)
```

```
$names
[1] "plddt"   "max_pae" "pae"     "ptm"     "iptm"
```

```
head(pae1$plddt)
```

```
[1] 91.44 93.75 94.19 93.44 95.62 89.81
```

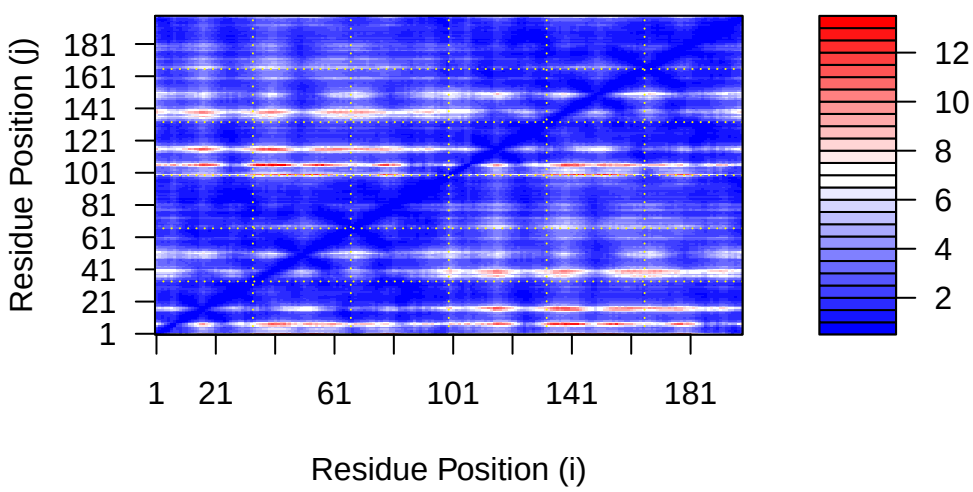
```r
pae1$max_pae
```

```
[1] 13.13281
```

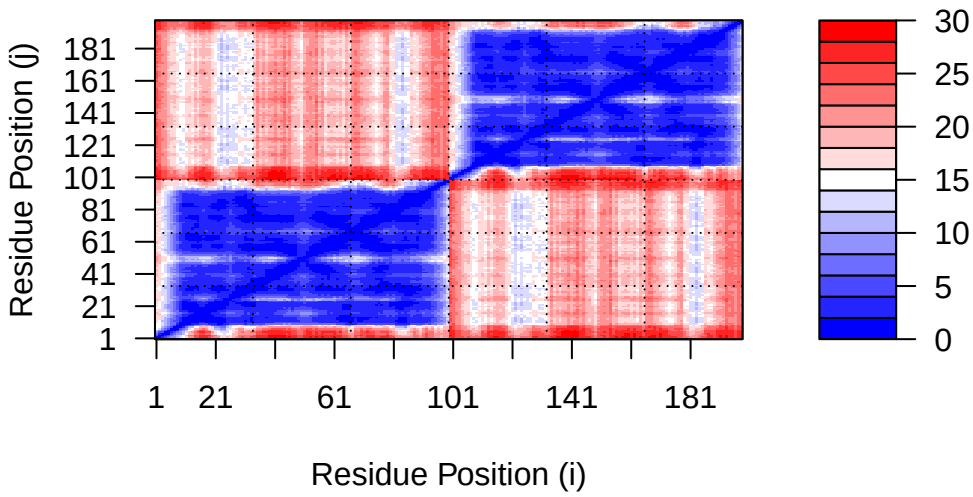
```r
pae5$max_pae
```

```
[1] 29.71875
```

```r
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)")
```




```r
plot.dmat(pae5$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```




```
aln_file <- list.files(path=results_dir,
                       pattern=".a3m$",
                        full.names = TRUE)
aln_file
```

```
[1] "hivpr_23119/hivpr_23119.a3m"
```

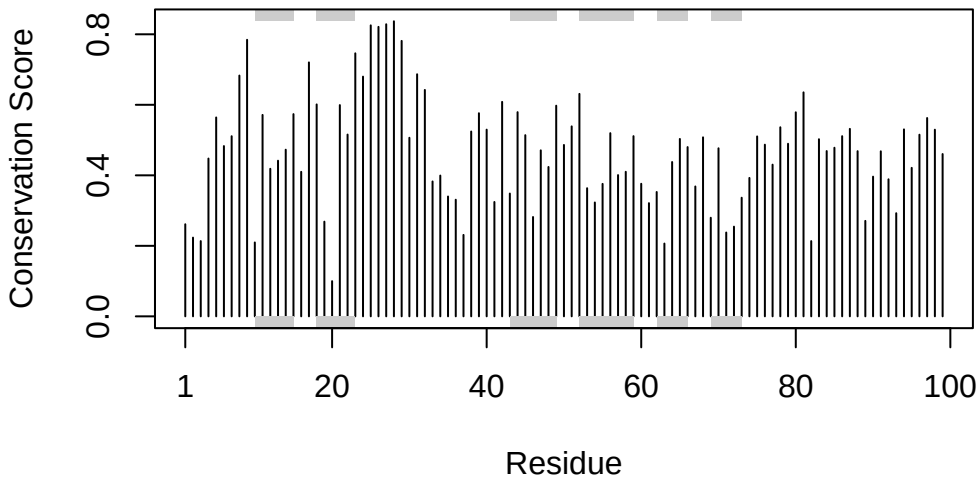
```
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

```
[1] " ** Duplicated sequence id's: 101 **"
[2] " ** Duplicated sequence id's: 101 **"
```

```
dim(aln$ali)
```

```
[1] 5397  132
```

```
sim <- conserv(aln)

plotb3(sim[1:99], sse=trim.pdb(pdb, chain="A"),
       ylab="Conservation Score")
```




```
con <- consensus(aln, cutoff = 0.9)
con$seq
```

```
  [1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-" "-"
 [37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"
```

```
m1.pdb <- read.pdb(pdb_files[1])
occ <- vec2resno(c(sim[1:99], sim[1:99]), m1.pdb$atom$resno)
write.pdb(m1.pdb, o=occ, file="m1_conserv.pdb")
```

Session Info

```
sessionInfo()
```

```
R version 4.5.2 (2025-10-31)
Platform: aarch64-apple-darwin20
Running under: macOS Tahoe 26.1

Matrix products: default
BLAS:   /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framewo
LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;  

locale:
[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8

time zone: America/Los_Angeles
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
[1] jsonlite_2.0.0  pheatmap_1.0.13 bio3d_2.4-5

loaded via a namespace (and not attached):
 [1] crayon_1.5.3        cli_3.6.5           knitr_1.51
 [4] rlang_1.1.7         xfun_0.55           otel_0.2.0
 [7] generics_0.1.4      glue_1.8.0          S4Vectors_0.48.0
[10] Biostrings_2.78.0   htmltools_0.5.9     stats4_4.5.2
[13] scales_1.4.0        rmarkdown_2.30      grid_4.5.2
[16] Seqinfo_1.0.0       evaluate_1.0.5      fastmap_1.2.0
[19] lifecycle_1.0.5     yaml_2.3.12         IRanges_2.44.0
[22] compiler_4.5.2      RColorBrewer_1.1-3  Rcpp_1.1.1
[25] XVector_0.50.0      farver_2.1.2        digest_0.6.39
[28] R6_2.6.1            parallel_4.5.2      tools_4.5.2
[31] gtable_0.3.6        msa_1.42.0          BiocGenerics_0.56.0
```